

Yonatan Tesfagiorgis

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SUMMARY

Software Engineer with experience building full-stack web applications, backend APIs, and AI-powered software solutions using Python, JavaScript, React, Flask, FastAPI, and SQL. Experienced in developing scalable systems, REST APIs, cloud deployment, and applying software engineering best practices.

EDUCATION

University of Houston

B.S. Computer Science

May 2026

Texas State Technical College

A.S. Cybersecurity

July 2019

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, PHP, HTML/CSS, TypeScript

Frontend: React.js, HTML5, Vite

Databases: MySQL, SQL Server, PostgreSQL, MongoDB

AI/ML: PyTorch, Hugging Face, LangChain, NLP

Backend: Flask, FastAPI, Node.js, REST APIs, Express.js

Cloud/DevOps: Docker, AWS, Azure, Git, CI/CD

TECHNICAL EXPERIENCE

Full-Stack AI Document Search Platform

January 2026 – March 2026

- Designed and deployed an end-to-end Retrieval-Augmented Generation (RAG) pipeline using LangChain, FAISS, Hugging Face, and Streamlit for conversational PDF querying across 50+ PDF documents.
- Implemented semantic chunking, vector embeddings, and retrieval workflows to process and index 10,000+ document chunks for efficient information retrieval.
- Built a FAISS-based semantic search engine supporting real-time retrieval from 10,000+ vector embeddings.
- Developed and deployed an interactive AI chatbot application on Hugging Face Spaces with support for 100+ document-based test queries.

NLP Sentiment Retrieval System

August 2025 – December 2025

- Built an NLP pipeline for aspect-based sentiment analysis using 10,000+ Amazon product reviews.
- Processed and transformed large-scale text data using tokenization, machine learning classifiers, and transformer-based embeddings.
- Implemented semantic similarity retrieval using sentence transformers and cosine similarity across 10,000+ text samples.
- Improved retrieval precision compared to keyword-based sentiment methods.

Full-Stack Volunteer Management Platform

June 2025 – August 2025

- Developed a full-stack volunteer management platform using Flask, React, and MySQL with REST API integration supporting 500+ volunteer records.
- Built dashboards, scheduling workflows, and automated PDF/CSV reporting features.
- Optimized database queries and backend services for efficient user and event management.

LEADERSHIP EXPERIENCE

Cougar CS Organization – University of Houston

January 2024 – May 2026

- Led C++ workshops and technical problem-solving sessions for workshop participants.
- Mentored peers in programming fundamentals, debugging techniques, and algorithmic thinking.